Results pack Q2: Cumulative fluid balance and acute kidney injury

FLUID-ICU is a SIMULATED dataset made for teaching. No real patients. Neudata | Foundations of Clinical Research, Level 1

SIMULATED DATA: NOT REAL PATIENTS. FLUID-ICU was generated by computer for this course. Write your manuscript as a training exercise and state clearly that the data are simulated.

1. The question

In adults with sepsis in the ICU, is a higher cumulative fluid balance by day 3 associated with acute kidney injury?

2. Design and data

- Design: multicentre prospective cohort study (observational).
- Setting: 4 ICUs (ICU A-D), consecutive adults with sepsis, enrolment January 2023 to December 2024.
- Exposure: cumulative fluid balance at day 3 in ml/kg, analysed per 10 ml/kg (continuous); secondary: positive vs negative balance.
- Outcome: acute kidney injury (KDIGO stage >= 1, maximum stage during ICU days 1-7; aki_any). Secondary: AKI stage 2-3; renal replacement therapy (descriptive).
- Analysis sample: 618 patients with a day-3 balance; adjusted models 527 complete cases.
- Pre-specified confounders (SAP): age, SOFA score, lactate, vasopressor, mechanical ventilation, chronic kidney disease.
- Software: R 4.6 (packages gtsummary, survival, mice). Scripts: Course/Data/R/.

Participant flow: 620 patients enrolled -> 2 without a day-3 fluid balance (weight or output missing after cleaning) -> 618 in the exposure analysis -> 91 without lactate (or age) -> 527 in the complete-case adjusted model.

3. Table 1: who was studied

Table 1. Baseline characteristics by fluid-balance group

Characteristic	Overall N = 618[1]	Negative balance N = 245[1]	Positive balance N = 373[1]
Age, years	62.6 (15.2)	64.6 (15.0)	61.4 (15.2)
Missing	1	1	0
Sex			
Female	243 (39%)	102 (42%)	141 (38%)
Male	375 (61%)	143 (58%)	232 (62%)
Weight, kg	64.7 (11.9)	65.1 (12.4)	64.4 (11.6)
Diabetes	158 (26%)	70 (29%)	88 (24%)
Chronic kidney disease	64 (10%)	28 (11%)	36 (9.7%)
Heart failure	97 (16%)	54 (22%)	43 (12%)
Hypertension	282 (46%)	124 (51%)	158 (42%)
Source of infection			
Respiratory	267 (43%)	101 (41%)	166 (45%)

Characteristic	Overall N = 618[1]	Negative balance N = 245[1]	Positive balance N = 373[1]
Urinary	126 (20%)	54 (22%)	72 (19%)
Abdominal	106 (17%)	46 (19%)	60 (16%)
Skin/soft tissue	57 (9.2%)	18 (7.3%)	39 (10%)
Other	62 (10%)	26 (11%)	36 (9.7%)
SOFA score	7 (5, 9)	6 (4, 8)	8 (6, 10)
APACHE II score	19 (15, 24)	17 (12, 21)	20 (17, 25)
Missing	6	3	3
Lactate, mmol/L	2.6 (1.7, 3.7)	2.2 (1.4, 3.0)	2.9 (1.9, 4.0)
Missing	91	50	41
Mean arterial pressure, mmHg	73 (66, 80)	76 (70, 84)	71 (65, 78)
Missing	4	1	3
Vasopressor on admission	266 (43%)	61 (25%)	205 (55%)
Mechanical ventilation	313 (51%)	97 (40%)	216 (58%)

[1]Mean (SD); n (%); Median (Q1, Q3)

Mean (SD); n (%); median (Q1, Q3). Missing = number of patients without that value.

4. Main results

- AKI (any stage) occurred in 302/620 (48.7%) of all patients; RRT in 79/620 (12.7%).
- AKI: positive balance 230/373 (61.7%) vs negative balance 71/245 (29.0%).
- AKI rose with each balance category: <= 0: 29%; >0 to 20: 54%; >20 to 40: 63%; > 40: 75%.
- Crude OR per 10 ml/kg more balance: 1.37 (95% CI 1.27 to 1.47).
- Adjusted OR per 10 ml/kg: **1.27 (95% CI 1.16 to 1.40)**, p <0.001.
- Positive vs negative balance: crude OR 3.94 (95% CI 2.80 to 5.60); adjusted OR 2.70 (95% CI 1.76 to 4.15).

Table 2. Maximum KDIGO AKI stage by fluid-balance group

AKI stage	Negative balance	Positive balance
KDIGO stage 0	174 (71%)	143 (38%)
KDIGO stage 1	29 (12%)	68 (18%)
KDIGO stage 2	24 (10%)	54 (14%)
KDIGO stage 3	18 (7%)	108 (29%)

Table 3. AKI by category of cumulative fluid balance at day 3

Cumulative balance day 3, ml/kg	n	Any AKI, n (%)	AKI stage 2-3, %	RRT, %
<= 0	245	71 (29%)	17.1	4.1
>0 to 20	166	90 (54%)	30.7	11.4

Cumulative balance day 3, ml/kg	n	Any AKI, n (%)	AKI stage 2-3, %	RRT, %
>20 to 40	130	82 (63%)	52.3	21.5
> 40	77	58 (75%)	55.8	28.6

Table 4. Logistic regression for AKI (adjusted for age, SOFA, lactate, vasopressor, ventilation, CKD)

Model	n	OR (95% CI)	p
Balance per 10 ml/kg, crude	618	1.37 (1.27 to 1.47)	<0.001
Balance per 10 ml/kg, adjusted	527	1.27 (1.16 to 1.40)	<0.001
Positive vs negative balance, crude	618	3.94 (2.80 to 5.60)	<0.001
Positive vs negative balance, adjusted	527	2.70 (1.76 to 4.15)	<0.001
Balance per 10 ml/kg, adjusted (outcome AKI stage 2-3)	527	1.23 (1.12 to 1.35)	<0.001

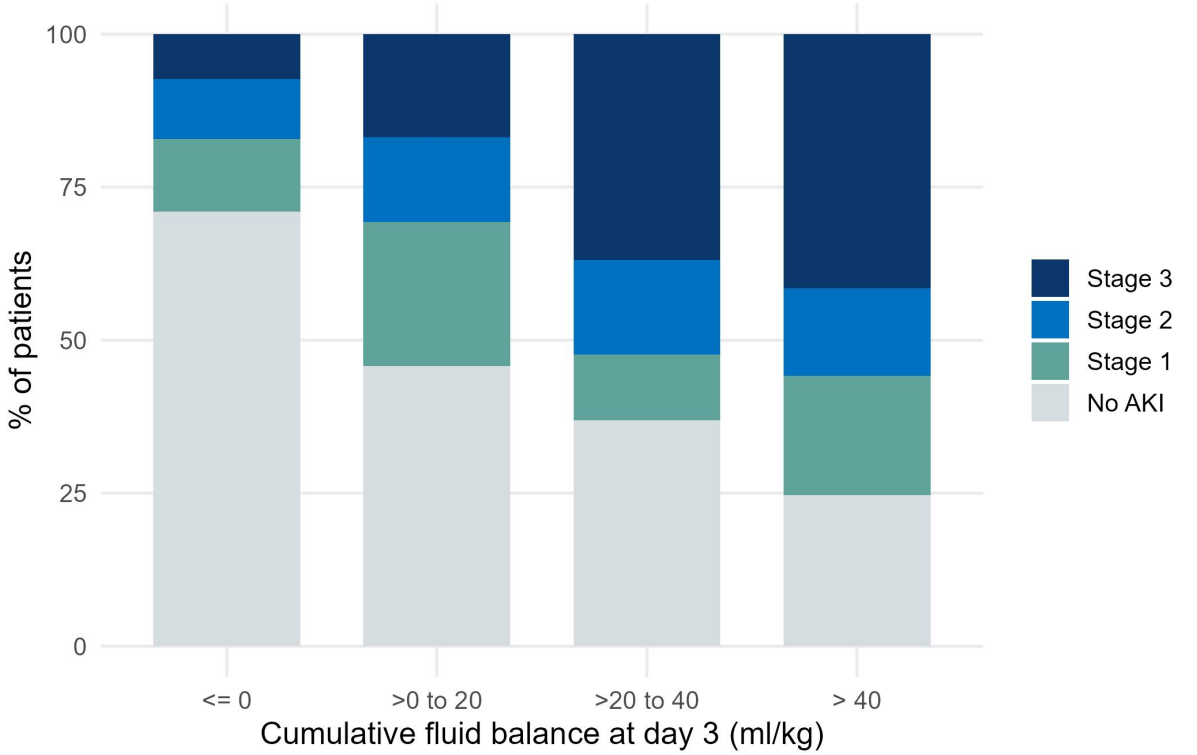


Figure 1. Maximum AKI stage by category of cumulative fluid balance

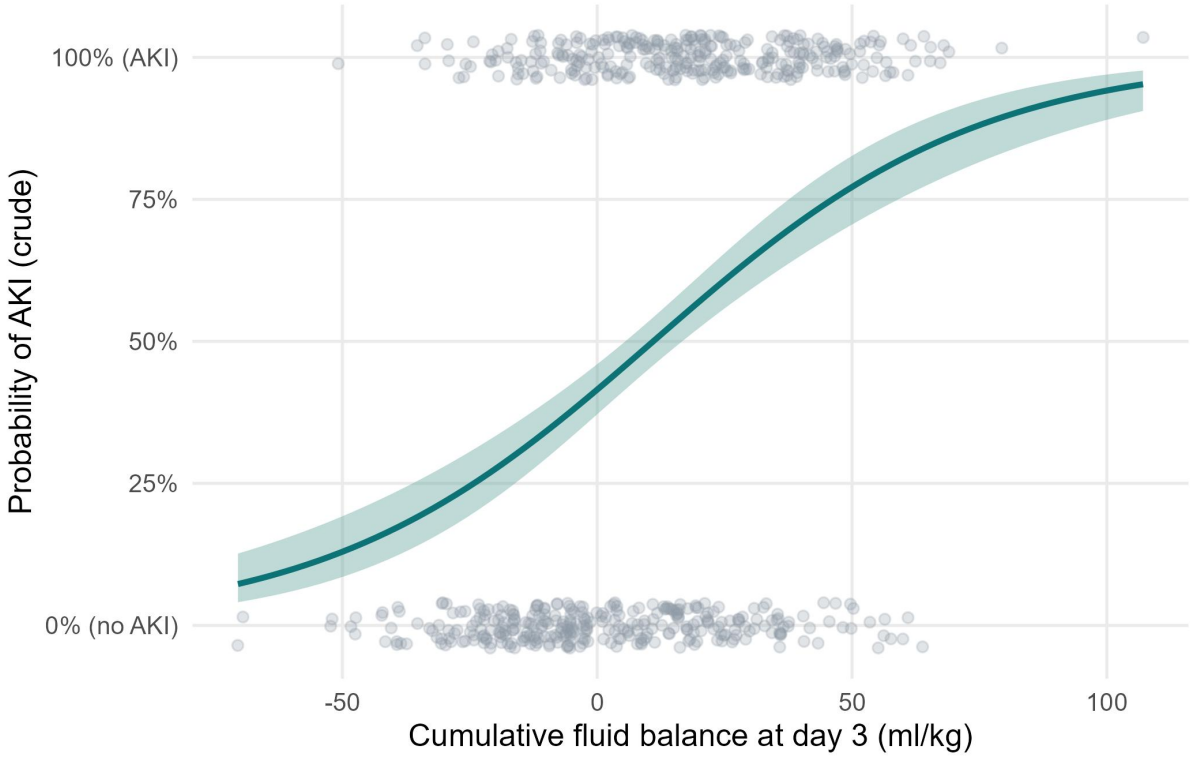


Figure 2. Crude probability of AKI across the range of fluid balance (logistic curve, 95% band)

5. What this means (plain language)

- AKI was more common the more positive the fluid balance: a clear 'dose-response' pattern.
- After adjustment, each extra 10 ml/kg (about 650 ml for a 65 kg adult) was associated with about 27% higher odds of AKI.
- Using the continuous balance keeps more information than positive/negative and shows the gradient.

6. What this does NOT mean

- It does NOT show that fluid damages the kidney. The arrow may point the other way: AKI reduces urine output, and low output makes the balance positive (reverse causation).
- We do not know whether AKI started before or after the fluid accumulated. The data record the maximum stage over days 1-7.
- An OR of 1.27 per 10 ml/kg is not a risk: with AKI this common (about 49%), odds ratios overstate risk ratios.

7. Limitations to discuss (hints)

- Timing: exposure (days 1-3) and outcome (days 1-7) overlap.
- Residual confounding by severity and nephrotoxic drugs (not recorded).
- KDIGO staging may be based on creatinine, which fluid dilutes, so AKI may be under-diagnosed in patients with large positive balances.
- Complete-case analysis (lactate missing in 15%).

Which STROBE items this pack helps you write

- Item 6-7: participants and variable definitions (KDIGO)
- Item 12: methods incl. continuous exposure scaling (per 10 ml/kg)
- Item 14: descriptive data (Table 1)
- Item 15: outcome data (Tables 2-3)
- Item 16: crude and adjusted ORs with 95% CI; category boundaries reported (Table 3-4)
- Item 19: limitations (reverse causation and timing)

- Item 17: secondary outcome AKI stage 2-3